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Introduction

What this book is about

What exactly constitutes a field like scientific psychology? Is it constituted by its most innovative and influential contributors; by the scientific findings that it has produced; by the theories it has elaborated; by its concepts, techniques, or professional associations? Obviously, all this and more goes into the making of a field, but most of us would probably see some of these components as playing a more essential role than others. Even if we refuse to commit ourselves explicitly we are likely to imply that certain components define the field more effectively than others by the way we organize our knowledge. For example, in the systematic presentation of information derived from the field of psychology or one of its parts, the material is most commonly organized in terms of prominent contributors, important findings, or influential theories. A perhaps unintended message of such communications is that psychology *is* its theories, *is* its findings, or *is* its individual contributors.

The way in which we organize a field will determine the way we organize its history. If we see the field of psychology as essentially an aggregate of individual contributors, we are likely to treat the history of the subject in terms of a succession of prominent figures. If psychology is its theories or its findings, then its history will become a history of psychological theories or psychological findings. Our organization of the history of the field will also serve as a subtle justification of the way we have characterized the field in the present.

Most psychologists have been taught to characterize their own scientific activity in terms of a framework that is derived from nineteenth-century physical science. They see themselves as individual investigators who seek to accumulate facts about some aspect of nature by the use of appropriate

hypotheses and techniques. When they describe the historical development of their field they are apt to do so in much the same terms, representing it as a succession of individual contributors who accumulated “findings” on the basis of progressively refined hypotheses and increasingly sophisticated instrumentation. Of course, the sad truth sometimes forces a departure from this framework, but the framework continues to operate all the same.

To put it very simply, this book is about some crucial elements that are missing in this framework. What is missing is the recognition of the socially constructed nature of psychological knowledge. The received view is based on a model of science that is reminiscent of the tale of Sleeping Beauty: The objects with which psychological science deals are all present in nature fully formed, and all that the prince-investigator has to do is to find them and awaken them with the magic kiss of his research. But in truth scientific psychology does not deal in natural objects. It deals in test scores, rating scales, response distributions, serial lists, and innumerable other items that the investigator does not just find but constructs with great care. Whatever guesses are made about the natural world are totally constrained by this world of artifacts.¹ The same holds true for the immediate human sources of the psychologist’s information. The psychologist’s interaction with such sources takes place within a well-regulated social role system, and such roles as that of experimental subject or of client in therapy are the direct result of the psychologist’s intervention.

In talking about a field like scientific psychology we are talking about a domain of constructions. The sentences in its textbooks, the tables and figures in its research reports, the patterned activity in its laboratories, these are first of all products of human construction, whatever else they may be as well. Although this seems quite obvious, certain implications are usually evaded. If the world of scientific psychology is a constructed world, then the key to understanding its historical development would seem to lie in those constructive activities that produced it. But this insight has not guided many historical studies. In the past the effects of a naive empiricism may have assigned an essentially passive role to investigators, as though they merely had to observe or register what went on outside them. But this is no longer a popular position.

In more recent times the well-known contrast between “context of discovery” and “context of justification” gave expression to a pervasive tendency to relegate the necessary subjective component in scientific activity to a mysterious underworld that was not susceptible to logical analysis.² So there grew up a strange duality in the historiography of fields like psychology, where one kind of historical review would restrict itself to the logical succession of hypotheses and evidence while the second kind would describe the personal lives of those individuals who were the authors of the hypotheses and the producers of the evidence. Whereas the first kind

of review implied that scientific progress was an affair governed by purely rational considerations, the second kind picked up the irrational component and located it in the personal quirks and accidental events that characterized the lives of historically important contributors. The two approaches were the product of a tacit consensus about the fundamental split between two components of scientific activity: a rational transindividual component, of primary importance in the context of justification; and an irrational individual component, important in the context of discovery.

What is missing from this account is any appreciation of the fundamentally social nature of scientific activity.³ What unites individual contributors is not simply their common possession of the same logical faculties and their common confrontation of the same external nature. Their social bonds are a lot more complex than that. They are related by ties of loyalty, power, and conflict. They share interests as well as logical faculties, and they occupy positions in wider social structures. In this social world of science the neat distinction between rational and irrational components crumbles. The fundamental issue in research is not whether the lone investigator can verify his hypotheses in the privacy of his laboratory but whether he can establish his contribution as part of the canon of scientific knowledge in his field.⁴ In other words, the issue is one of consensus, and consensus is not entirely a matter of logic. It involves prior agreements about what is to count as admissible evidence and shared commitments to certain goals. It involves vested interests and unexamined biases.

Once we recognize the essentially social nature of scientific activity, we are compelled to see both the "context of discovery" and the "context of justification" in a different light. The context of discovery is in fact a context of construction, of theories, of instruments, and also of evidence. For the data that appear in the pages of psychological journals are no less a product of the constructive ingenuity of their authors than are the instruments and the theoretical hypotheses; they are not raw facts of nature but elaborately constructed artifacts.⁵ However, these artifacts are constructed according to explicit rational schemes accepted within a certain community of investigators.⁶ That is why there is a real history of psychological research practice that is neither a series of narratives about famous psychologists nor an enumeration of their successive "findings." This history involves the changes that have taken place in the constructive schemes that psychologists have used in the production of those objects that form the accepted content of their discipline.

But what is meant by these "constructive schemes"? In the first place, such schemes are not just cognitive frameworks for the *interpretations* of empirical data but involve practical rules for the *production* of such data. It is true that general concepts and theories also function as constructive schemes that give a particular meaning to the objects in which the discipline deals. But such interpretive schemes are found in purely speculative psy-

chologies as well as in empirical psychology. Psychology underwent a fundamental change toward the end of the nineteenth century when its practitioners became decisively committed to specific practical methods of data production. The application of these methods became the special characteristic of the field and distinguished it from everyday psychology as well as from its own intellectual predecessors.⁷ By the use of these practical methods, modern psychology created a new world of psychological objects that increasingly defined the field and to which any purely theoretical developments were forced to accommodate.

This emphasis on practical constructive schemes does not however lead simply to a history of psychological methods in the conventional sense. The difference lies in the way one conceives of method. Conventionally, empirical methods in psychology are conceived simply as tools for the achievement of certain technical goals. Thus, they need be evaluated only in terms of a logic of means and ends, where the ends are taken as existing independently of and prior to the means. For instance, we may choose the measurement of a particular psychological quality, such as intelligence, as an end, and then trace the effectiveness of various instruments for achieving this end. Or we may take it as an end that we want to assess the simultaneous contribution of a number of factors to psychological effects and then attribute the growing popularity of certain statistical techniques to their superior effectiveness in achieving this preexisting goal. It is possible to trace the history of psychological methodology in terms of such a purely instrumentalist framework, but that is not what is being attempted here.

The concept of investigative practice is wider than the concept of methodology. As conventionally understood, the latter involves an abstraction of certain rational and technical aspects of investigative practice from all the other aspects. The practice of investigators is treated as though it consisted only of logical and technical operations performed by independent individual investigators on bits of the natural world. Left out is the fact that investigative practice is very much a social practice, in the sense that the individual investigator acts within a framework determined by the potential consumers of the products of his or her research and by the traditions of acceptable practice prevailing in the field.⁸ Moreover, the goals and knowledge interests that guide this practice depend on the social context within which investigators work. Finally, in psychological research there is the additional important consideration that the investigator is not the only human participant whose actions are necessary for the practice of investigation to proceed. Unless the psychologist works with animals, he or she will also require the collaboration of human sources of data without whose contribution there would of course be nothing to report.

The notion of investigative practice then involves the social dimensions of research activity as much as the logical ones. The latter are recognized

as being embedded in a social matrix that includes such factors as the pattern of social relations among investigators and their subjects, the norms of appropriate practice in the relevant research community, the kinds of knowledge interests that prevail at different times and places, and the relations of the research community with the broader social context that sustains it.

As long as we limit our conception of psychological research practice to its purely rational aspects, we will be inclined to think of the history of that practice solely in terms of technical progress. The norms of good scientific practice will be seen as belonging to an unchangeable trans-historical realm where eternal rational principles rule. All that is left for worldly history is the narrative of how clever investigators came to apply these eternal principles with ever greater concrete effectiveness to larger and larger bodies of knowledge. However, if we refuse to perform this rationalist reduction, we will find that in the history of psychological research practice the most significant changes were changes in the ends rather than improvements in the means.

Although methodological rationalism has adherents outside psychology, the doctrine has long played a particularly important role for this discipline. In the more established fields of natural science, where elaborate deductive procedures are part and parcel of theoretical discourse, theory as much as method was regarded as the repository of scientific reason. But in psychology theoretical constructions have seldom been marvels of logical sophistication, and pessimism about the likelihood of reaching rational consensus on the basis of theory has long been widespread. It is generally accepted that there will be controversy about theoretical fundamentals and that personal, cultural, and historical factors play important roles in the elaboration and acceptance of psychological theory. But this state of affairs is hardly compatible with the claims by the discipline for the objectivity of its insights into human behavior. Therefore, such claims in psychology have depended almost completely on the rational virtues of its methodology. It was only because of the logical-technical features of its investigative practice that psychology could give some plausibility to its claims for scientific status.

Investigative practice therefore constitutes an area of considerable anxiety within the discipline of psychology.⁹ Concern with questions of methodological orthodoxy often takes the place of concern about theoretical orthodoxy when research or its results are discussed and evaluated. These preoccupations with the purity of method frequently deteriorate to a kind of method fetishism or "methodolatry."¹⁰ From this point of view there may be something distinctly subversive about the suggestion that the sphere of methodology is not a realm of pure reason but an area of human social activity governed by mundane circumstances like any other social activity.

Nevertheless, the consequences of this suggestion should be explored, for not to do so exposes one to all the risks entailed by a naive and self-deluded style of scientific practice.

Once we restore the abstraction of a purely rational methodology to the broader context of investigative practice, it becomes possible to see it as the primary medium through which social forces have shaped the production of the objects of scientific psychology. But this general pattern can only be fully appreciated when studied in its historical development. Before we can undertake this task, however, we require some clarification of the sociological and historiographic aspects of the topic.

The social generation of scientific knowledge

In recent years references to science as a social activity have become commonplace, and there is a rapidly growing field devoted to the social study of science. Whereas the earlier sociology of science was more a sociology of scientists and exempted the content of science from its purview, more recent developments in this field have pointed in the direction of a genuine sociology of scientific knowledge.¹¹ In other words, scientific knowledge has increasingly been seen as a product of certain quite specific social processes, and many of its features have been studied in terms of their dependence on these processes. With a few notable exceptions,¹² psychological knowledge has been largely exempt from this process. This may be due to the existence of two large bodies of opinion: For one the social dependence of psychological knowledge is too obvious to be worth studying, whereas for the other the whole topic is too threatening to psychology's hard-won scientific status to be taken seriously. The fact is, however, that psychology's social practices of investigation are too tied up with its claims to scientific status to be ignored in that context. Still, our understanding of the social determination of psychological knowledge must remain rather abstract until we understand the crucial mediating role played by investigative practice.

It may be helpful to think of the social contexts of investigative practice in terms of three concentric circles of varying radius. (Figure 1.1) The innermost circle represents the immediate social situation in which the information that will later be transformed into research data and scientific psychological knowledge is generated. The next circle represents the research community by which the product of the investigation must be accepted if it is to count as scientific psychological knowledge. The outer circle stands for the wider social context within which the research community is embedded – the sources of research support, professional institutions, potential external consumers of knowledge products and skills, representatives of prestigious disciplines, and so forth.

The image of concentric circles is of course meant to represent the idea

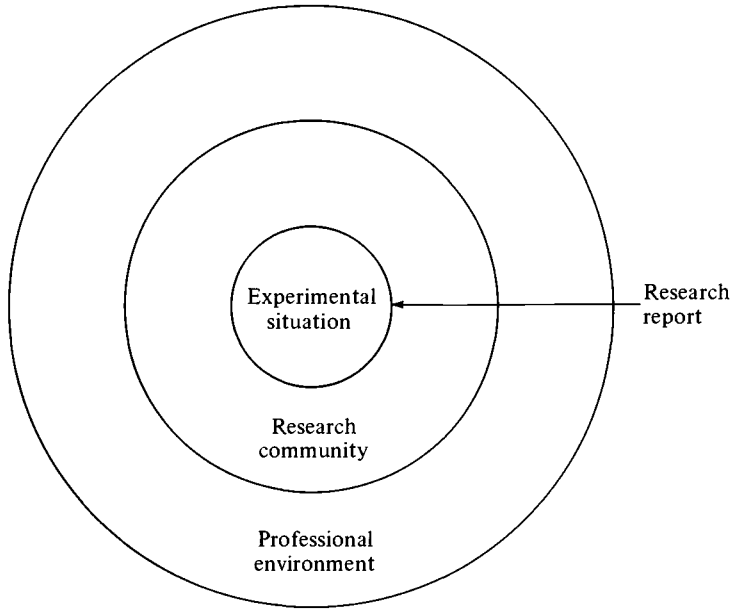


Figure 1.1. Social contexts of investigative practice

that both the immediate investigative situation and the research community are to be regarded as embedded in social relationships that extend beyond them. Similarly, a truly comprehensive account would have to add further circles beyond the three discussed here to represent even broader cultural, political, and economic contexts. But because our concern is specifically with investigative practice, we will limit ourselves to the inner three circles, considering in turn some of the more important questions that arise within each.

At the innermost level we have to deal with what are essentially face-to-face situations. Before the whole process of producing material that counts as scientific psychological knowledge can begin, some individuals have to get together to generate the information that forms the starting point for this process. In coming together for this purpose they set up certain social situations with a distinct character – the situations of mental testing, psychological experimentation, or interviewing, to mention the most obvious ones. In the case of the first two, which are by far the most important ones in psychology, the social interaction of the participants is partly mediated by various kinds of hardware and software, mental-test materials, and laboratory apparatus in particular. This has permitted investigators to overlook the social character of all these investigative situations and to pretend that their subjects reacted only to the dead materials presented to them and not to those who did the presenting. Nevertheless,

it is an elementary truth that the gathering of psychologically relevant information requires the active social participation of individuals who will act as the source of this information. Whatever else it may be, the psychological experiment or test is therefore a social situation and as such it must share the characteristics that are found in all social situations.

Of course, there are marginal cases where the social status of the investigative situation is less straightforward, such as the use of documentary sources or the case where individuals experiment on themselves. But neither of these has accounted for more than a tiny fraction of psychological research over the past century. Their rarity only serves to emphasize the inherently social nature of most psychological investigations.

Until relatively recently the total blindness of psychological investigators to the social features of their investigative situations constituted one of the most characteristic features of their research practice.¹³ However, the recognition of so-called experimenter expectancy effects and demand characteristics has now become commonplace, and there is a large empirically oriented literature on "the social psychology of the psychological experiment."¹⁴ Although this undoubtedly represents an advance on psychologists' traditional naiveté in regard to such issues, the limitations of this literature are more remarkable than its achievements from the viewpoint of furthering an understanding of the social process of generating psychological knowledge.

In the first place, much of the literature on social effects in psychological experiments is devoted simply to demonstrating the mere existence of such effects without being directed at furthering an understanding of the social processes involved in psychological investigation.¹⁵ Such conceptualization of social process as does occur in this work is usually confined to social *psychological* rather than sociological categories and therefore does not address the fact that the responses of individuals to experimental situations take place in the context of a miniature social system in which both experimenters and their subjects participate. Moreover, there is a pervasive tendency to relegate the social aspects of psychological experimentation to the status of "artifacts" or, in other words, disturbances of the process of research that do not belong to its essential nature.¹⁶ Thus the rational abstraction of a purely logical, asocial, and ahistorical research process remains inviolate and apparently beyond the reach of empirical correction. Accordingly, in this research program the preferred method for studying the social aspects of experimentation is the experimental method itself, an enterprise whose results are difficult to interpret without getting mired in an infinite regress.¹⁷

Such problems mostly derive from a single source, namely, the pretense that psychological experiments are not in principle different from experiments in the natural sciences.¹⁸ Because in the latter case experimenters are able to treat whatever they are investigating purely as a natural object,

it is believed that human data sources must be treated in the same way if psychology is to function as a proper experimental science.¹⁹ But human subjects in psychological experiments are in fact unable to behave simply as natural objects. Even if they try to do so, which depends entirely on their appraisal of the social situation they are in, they negate this fictional goal in the very act of trying to reach it, because such efforts represent an exercise of their social agency.²⁰ Psychological experiments are therefore different in principle from experiments in physics because the experimenter and the human data source must necessarily be engaged in a social relationship. This is no “artifact” but one of the essential preconditions of having a viable experimental situation. Consequently, experimental results in psychology will always be codetermined by the social relations between experimenters and subjects. The precise significance of this factor may be variable, but that it is always present is not a matter open to doubt.

Of course, the old notion that the knowledge produced by experiments depends only on the interaction of individual experimenters with the materials of nature is false even in the case of the true natural sciences. We now understand a great deal about how the kind of knowledge gain that results from an experiment depends quite crucially on the interaction of scientists with each other.²¹ But this is a level of social influence that psychological research shares with research in other experimental sciences. Beyond this there is however a level of essential social interaction peculiar to psychology experiments – the interaction of experimenters with the human sources of their data. It is to be hoped that after a century of lusty life, experimental psychology now feels sufficiently secure to face the fact that there may be essential differences between itself and the older experimental sciences. The crude doctrine that relegated everything not cast in the image of experimental physics to the category of the “mystical” is being increasingly treated as a historical curiosity.

Finding out more about the social relationships that exist between experimenters and their human subjects is surely better than pretending they are not there or are of no importance. Now, there are different ways of pursuing such a goal. The experimental way, which has already been alluded to, tends to be limited to investigating certain *social psychological* factors that operate in psychological experiments.²² But psychological factors like the experimenter’s expectancies or the subject’s “evaluation apprehension” operate within a certain social framework that has to be taken for granted in such studies. This framework is provided by the traditions and conventions of psychological experimentation that have developed over the years and are now well understood by all experimenters and by most of their subjects. In those societies in which it is practiced on any scale, the psychological experiment has become a social institution recognized by most people with a certain level of education. As in all social institutions the interaction of the participants is constrained by institutional

patterns that prescribe what is expected and permitted for each participant. The successful conduct of psychological experiments depends on the willingness of all participants to abide by these rules, and that of course can only happen if they have the appropriate level of previous knowledge to understand the rules. We know that in cross-cultural research this cannot always be taken for granted.

But before we discuss this as a third-world problem we might remember that hardly more than a century ago the institution of the psychological experiment was as unknown everywhere as it might now be in parts of the third world. It is legitimate to ask how this social construction ever arose in the first place and how it ended up taking the form we are so ready to take for granted today. Like all social institutions, the psychological experiment not only has a certain social structure that can be analyzed but a history that can be traced. Ultimately, this institution is part of the history of those societies that produced it and can be expected to bear the marks of its origins. Thus, if we wish to improve our understanding of the accepted social framework within which individual participants in psychological experiments have to function, we will have to adopt a historical rather than an experimental approach. For neither experimenters nor their subjects enter the investigative situation as social blanks to be programmed in an arbitrary manner. Both are the products of a distinct historical development that has left a heavy sediment of blind faith and unquestioned tradition. It is precisely this historical development that we will attempt to trace in the present volume.

Although studies of the social framework for the interaction of investigators and their subjects are virtually nonexistent, we move onto relatively more familiar ground when we consider the two outer circles that determine the social practice of research. In their interaction with each other and with a wider social context, psychological investigators face problems that are not different in principle from the social problems faced by other groups of scientists. From this perspective we recognize the social practice of scientific psychological investigation as involving a certain organization of work for the purpose of generating a certain kind of product that is identified as psychological knowledge. This identification of the product is accomplished by what amounts to a system of certification on the part of recognized authorities working within an established institutional framework.²³ The kind of knowledge the scientific worker is after is not private but public knowledge.²⁴ That means it cannot be achieved without acceptance by some kind of public. If the knowledge is to bear the stamp of science, the public by whom it is accepted must be a scientific public.

Psychological research is not something that is carried on by individual investigators working on their own account. Nor are its social aspects limited to the interaction between investigators and the human sources of their data. In designing their experiments and in preparing the cognitive

yield of their investigations for publication, researchers must be constantly mindful of the acceptability of their product to a particular public. That acceptability depends on whether the product measures up to currently prevailing standards of what constitutes scientific psychological knowledge. The fact that these standards have been thoroughly internalized by most investigators in the course of their professional socialization does not in any way affect the public status of these standards. What this means is that the scientific community is intimately involved in the social practice of scientific investigation.

As in the case of experimenter–subject interactions, there are two different levels on which the analysis of this aspect of social practice can be pursued. There is the social psychological level where one would be asking questions about how individual investigators adapt to prevailing standards of scientific knowledge and how they negotiate different interpretations of these standards. But at this level the existence of the standards themselves would have to be taken for granted. The systemic or historical level of analysis would treat precisely this existence of current standards as problematic and would attempt to trace their historical origin and development. This level of analysis will be pursued here.

Some historiographic considerations

The fact that currently dominant standards of what is to count as scientific psychological knowledge have not always existed and were not always dominant is one that is universally recognized. Every well-trained research worker in the field knows that the very idea of a scientific psychology only arose in relatively recent times and that its practical enactment belongs to an even more recent period. It is also well known that subsequent to the founding of experimental psychology there was a major controversy about the standards of what was to constitute scientific psychological knowledge. This controversy is usually represented as one between introspectionists and behaviorists, though, as we will see later, the story is more complex than that. What is worthy of note, however, is that an arch introspectionist like E. B. Titchener always justified his investigative practice in the name of *science* and denigrated the practice of his opponents as being not science but technology.²⁵ The more naive textbook treatments describe this controversy simply as a conflict between truth and error, the implication being that there can be only one conceivable true version of what constitutes scientific psychological knowledge. But if we rid ourselves of the irrational presumption that truth will necessarily vanquish error, and that therefore whatever achieves historical dominance must be truer,²⁶ this controversy becomes one between rival conceptions of what was to count as scientific psychological knowledge.

In studying historical changes in prevailing conceptions of psychological

knowledge, the question of the relative truth of these conceptions is not relevant. First of all, we are dealing with standards, criteria, and ideals rather than with matters of fact, and so the attribute of truth is simply an inappropriate one. The introspectionists' choice of a different kind of knowledge was a matter of preference, not a matter of error. Whether what they did was scientific depends on one's definition of science. If their definition turns out to be different from that of the behaviorists, this again is a question of preference, which can only be seen as a matter of being right or wrong if a particular definition of science is accepted as the only true one in some absolute, ahistorical sense.²⁷

But such ahistorical standards usually turn out to be the particular standards that happen to be currently popular. Elevating them to the status of final criteria therefore involves the patent illusion that historical development has reached a kind of culmination in the present and will now stop. The conservative implications of this position need no elaboration. When we attempt to trace historical changes in the social practice of investigation, we have to put questions of truth on one side. For here we are not dealing merely with technical means whose logical and practical adequacy can be assessed in terms of some accepted goal but with the goals themselves. Particular scientific communities are characterized by their commitment to certain scientific goals and this commitment determines the nature of their members' investigative practice.

What is certainly relevant in this context are the reasons for existing commitments to certain ideals of scientific practice and the reasons for changes in these commitments. These reasons are to be found in the common historical situation faced by members of a particular scientific community. Such communities never exist in a social vacuum, of course, but find themselves under the necessity of adopting positions in relation to other groups of investigators, to those who control the material resources for research, and to the general lay public. There may be other groups of investigators who are particularly well established and influential and who serve as models of successful practice. Those who ultimately control social resources must be persuaded to divert some of them to particular groups of investigators rather than to other purposes. At all times a newly emerging discipline like scientific psychology had to be careful to distinguish its cognitive product from the everyday knowledge of lay publics and from the rival claims of other disciplines. The requirements of potential consumers of a discipline's cognitive products will also pull their weight. All these and other similar factors help to shape the kinds of knowledge goals that prevail in a given field at a given time and therefore determine the dominant patterns of investigative practice.

This level of social determination can be thought of as constituting the outermost of the three concentric rings that were previously used to represent the multiple layers of the social practice of investigation. The in-

nermost ring represents the direct social interaction of the participants in the research situation, and the middle ring represents the interaction of members of the scientific community with each other. Although each circle makes its own specific contribution to the overall pattern of investigative practice, these contributions are not autonomous but derive their significance from their embeddedness in a wider social context.

Any attempt at studying the historical development of investigative practice faces certain practical problems. The most significant of these problems arises from the fact that the realm of practice involves a great deal of that "tacit knowledge" that has long been recognized as playing a crucial role in scientific investigation.²⁸ In fact, the tacit component tends to be much larger on the level of practice than on the level of theory. Aspiring investigators are typically socialized into the craft by following examples and may never be required to account to themselves or to others for choosing procedures that have the weight of tradition behind them. Moreover, the accounting that they are required to do is typically limited to purely technical considerations. It is precisely the social aspects of scientific practice that are systematically excluded from practitioners' discussions about methodology. In psychology, where some of these aspects can no longer be overlooked entirely, they are discussed as "artifacts" that may disturb the normal rational course of investigative practice.

In other words, investigators tend to be poor direct sources of information when it comes to certain aspects of their practice. This would not matter too much if we wanted to study contemporary practice because we might be able to observe investigators actually at work or question them at length about their actions.²⁹ But if we want to understand current practice in terms of its historical development, we are precluded from proceeding in this way. Documentary evidence, where it deals explicitly with problems of investigative practice, is typically limited to the purely technical aspects. In particular, textbook discussions of methodological issues tend to suffer from this limitation. They only become historically interesting when changes in the treatment of certain issues are compared over time, an approach that permits the emergence of some fundamental questions whose answers are taken for granted in any single textbook presentation. But only certain aspects of investigative practice ever receive extensive textbook treatment, and for the rest we must look for other sources of evidence.

Methodological prescriptions by prominent representatives of the field can sometimes be of value. However, such statements tend to belong to the ideology rather than to the practice of research. They tell us what their authors thought they were doing or what they wanted others to think they were doing, but they do not necessarily tell us what they actually did. The significance of such statements always needs to be assessed against independent evidence about actual practice. In any case, the statements of individuals, no matter how prominent, may have little relationship to gen-

eral practice, which is our ultimate interest. Maintaining a clear distinction between actual practice and discourse about practice is necessary because the latter involves the ways in which practitioners justify their practice to themselves and to others.³⁰ A form of social accounting, it provides only a very partial and often a distorted reflection of the practice as it might appear to a less partial observer with a broader perspective.

In regard to discourse about practice, it is difficult to escape the impression that its significance at different times in the history of modern psychology has varied widely. Such discourse was obviously crucial in the early years of experimental psychology, when the field was trying to define itself and to differentiate itself from related fields. In dealing with this stage, an understanding of prevailing discourse about practice is as important as an analysis of actual practice. But after the end of World War I, at least in the United States, psychology seems to have entered a more settled period during which answers to most of the fundamental methodological issues were increasingly taken for granted and discussion became limited to technical issues. (Nearer our own day there was a revival of interest in the more fundamental issues, but this takes us beyond the historical scope of the present volume.) Accordingly, we will pay less attention to discourse about method than to an analysis of actual practice. For, as we will see, actual practice underwent considerable change even though this was not always adequately reflected in current discourse about practice.

Fortunately, the conventions of scientific practice provide us with a veritable mine of information about those aspects that are ignored or distorted in discussions about technique. The products of research practice consist of reports published in special media that have a certain standing in some relevant research community. Such products constitute a large part of the crucial public domain of science. As such they are preserved for posterity and are therefore available for inspection as historical documents long after they have served their original purpose.³¹ These published research reports in recognized scientific journals provide us with the historical record that enables us to trace significant changes in investigative practice over the years.

Given their potentially enormous informative value and their relatively easy accessibility, it is surprising that research reports have so rarely been subjected to systematic historical study.³² Part of the reason for this may lie in the tendency to treat such reports at face value even though many years may have elapsed since their original publication. But a research article can inform us about much more than the specific data its authors wanted to communicate to their contemporaries. By the time that experimental psychology emerged, the conventions governing scientific research publications had developed to the point where they prescribed that in principle enough information about procedures had to be supplied to enable other investigators to replicate the published study. Of course, pub-

lication practices often fell short of this ideal, and in psychology it was in any case unrealistic because the historically unique nature of each study was accentuated by the use of human data sources. Nevertheless, the conventions of scientific publication exerted sufficient influence to ensure the inclusion of a considerable amount of information that bears on the development of investigative practices. For example, psychological research reports always contain some direct information about the identity of the participants in the research study and enough indirect information to enable one to draw some conclusions about the kind of relationship that must have existed between investigators and their human subjects.

The systematic analysis of research articles published in the major journals of the field will therefore form an important source of information for our historical study of the development of research practice. Much of this information is obtained by analyzing aspects of the research reports to which their authors would have attached little significance at the time. What these authors were interested in was the communication of their findings in an acceptable form; what we are interested in are, first, the tacit, taken-for-granted aspects of their social practice that enabled them to produce something they recognized as “data” in the first place, and second, the means they used to make these data presentable.³³ Thus our analysis will attempt to probe behind the conventional facade of the research article to the kinds of practices of which it was the product.³⁴

To return once again to the representation of the social practice of investigation by three concentric rings, as developed in the previous section, the published research report can be thought of as existing on the boundary between the innermost and the middle ring. In one direction the research article reflects something of the social interaction among the participants in the research situation, but it is also directed at a particular scientific audience by which it hopes to be accepted. To be so accepted, it must present its information in such a way as to conform to certain standards of what constitutes acceptable knowledge in the field. An appropriate analysis of research articles can therefore provide us with some insight into the two levels of social construction that have gone into its production: first, the social structuring of the research situation in such a way as to yield a certain type of information, and second, the restructuring of this information to make it fit a certain prescribed model of what scientific knowledge in the field should look like.

But even if this kind of analysis were successful, it would yield at best a static snapshot view of investigative practices at a particular time and in a certain locale. In reality the practices we are interested in constitute a moving target in the sense that they are subject to historical change, partly because of conflicts within the scientific community, and partly because they are embedded in a wider context of historical change (the outermost of our three circles). Thus, we will have to extend our analysis over several

decades to get a sense of the trend of historical development of investigative practices. Because the sheer volume of research output makes it practically impossible to analyze every relevant published article, it will be necessary to sample publications at an interval of several years.

We will also have to limit the scope of our inquiry to one part of the history of modern psychology. There can be no question that that part will have to be the earlier rather than the later period in the development of the discipline. Not only would it be difficult to achieve an understanding of later historical developments without a knowledge of the preceding period, but there would be a real danger of missing some of the most significant issues. By the middle of the twentieth century psychological research practices had settled into a rather rigid mold that allowed for relatively little variation.³⁵ But during the first half-century of its existence the investigatory practices of the new discipline were still in flux and diverged widely from each other. The roots of later rigidity lie in this earlier period, so that an analysis of what happened in this period can contribute to an understanding of the more familiar patterns that followed.

These temporal limitations allow a more comprehensive coverage of the research literature in the field than would otherwise be practicable. Such coverage is important because of the systemic level of the projected analysis. The kinds of processes we will be dealing with here are essentially collective processes that characterize the common practices of many individuals. Individual contributions are of interest only insofar as they represent a significant general trend. Thus, the role of specific individuals becomes relevant to our analysis primarily at the beginning of our period when modern empirical psychology first emerges as an identified field of activity for a distinct group of practitioners. Very soon, however, certain traditions of investigative practice form within the new field and subsequent developments are due to collective responses to conflicts among rival traditions and to pressures from outside. Although we need to understand the contributions of a few particularly influential pioneers at the point of emergence of the new discipline, the formation of distinct research communities soon makes it more appropriate to adopt a different level of analysis.³⁶ In the main we will be interested in identifying the pattern of variation, both synchronic and diachronic, that characterized the investigative practices of the discipline as a whole, rather than in assessing the role played by specific individuals.

2

Historical roots of the psychological laboratory

The birth date of modern psychology is usually placed toward the end of 1879 when Wilhelm Wundt designated some space at the University of Leipzig to be used for the conduct of psychological experiments.¹ Of course, the date is arbitrary, as are all such birth dates for disciplines. This arbitrariness arises less from the rival claims of other locations or individuals than from the obvious fact that the birth of a discipline is not a singular event but a complex process extending over a considerable period of time. In the case of psychology the relevant period extends both before and after the magic date.

When psychology became an autonomous field of research it did not invent its concepts and problems out of the blue but took them over from already existing fields like philosophy and physiology. Similarly, the practical activities that came to be identified as methods of psychological research were anything but completely new inventions. They were more in the nature of adaptations of already existing practices to a somewhat different context. Those who laid the foundations of a new psychological research tradition were constrained at every step by the traditions of investigative practice with which their general cultural experience or their personal training had made them familiar. The best they could do was to modify these practices, sometimes in quite minor ways, to suit the new goals they had in mind.

When Wundt proceeded to give practical shape to his program for a new kind of psychology, he based himself on three traditions of investigative practice, two of them explicitly, the other implicitly. Explicitly, he based himself first on the specific form of experimental practice that had recently grown up in physiology. This provided him with certain material techniques and a certain way of asking research questions. Second, he proposed to apply these procedures to a different object of investigation than the objects